

8	(a)	(i)	Amount of substance Thermodynamic temperature.	B1
			(award 1 mark for “no of moles and temperature”).	B1
		(ii)	Molecules collide with one another in a haphazard manner hence they possess different kinetic energies at any time.	B1
		(iii)	No intermolecular forces in ideal gas, hence no potential energy	B1
		(iv)	Low pressure High temperature	B2
	(b)	(i)	ρ = density of gas $\langle c^2 \rangle$ = mean square speed	B1 B1
		(ii)	$p = \frac{1}{3} \rho \langle c^2 \rangle$ $\Rightarrow pV = \frac{1}{3} M \langle c^2 \rangle$ since $\rho = M/V$ For ideal gas, $pV = nRT$ Hence, $\frac{1}{3} M \langle c^2 \rangle = nRT$ Total KE = $\frac{1}{2} M \langle c^2 \rangle = \frac{3}{2} nRT$	M1 B1
		(iii)	1. Same because average K.E. is proportional to temperature 2. Different because the masses of the gases are different.	B2
		(c)	(i)	
			$pV = nRT$ $(6.0 \times 10^5)(0.10) = (10)(8.31)T$ $T = 722 \text{ K}$	M1 A1
		(ii)	The products of p and V are not constant. For example, the product of p and V is $5.0 \times 10^4 \text{ J}$ at B, but $6.0 \times 10^4 \text{ J}$ at A. (Accept calculation of temperature at B = 600 K)	B1
		(iii)	Work done in one cycle estimated from the area enclosed $= 8.8 \times 10^4 \text{ J}$ (Note the positive sign as net work is done by the gas)	M1 A1
		(iv)	In one cycle $\Delta U = q + W = 0 \Rightarrow q = 8.8 \times 10^4 \text{ J}$	A1
		(v)	Curve for process A $\rightarrow$ B gentler/enclosed area larger Hence W larger implies, q is larger.	B1 B1

9 (a) The Principle of Superposition states that when two or more waves of the B1